(a)

Formability classification

Feature Relative Importance (%)

0
10
20
30
40

$IR_{B+B'}$
$IR_{A/B}$
IR_{B+X1}
B
$\chi_{B+B'}$
B'
$\chi_{B'+X1}$

(b)

Bandgap classification

Feature Relative Importance (%)

0
5
10
15
20
25

B
χ_B
IR_B
$IR_{B'}$
$IR_{B/B'}$
B'
$\chi_{B-B'}$
$IR_{B+B'}$

(c)

Bandgap regression

Feature Relative Importance (%)

0
10
20
30
40
50
60

$\chi_{B+B'}$
$IR_{B'-X1}$
χ_{B-X1}
$\chi_{B'-X1}$
$\chi_{B/X1}$
$\chi_{B/B'}$
B'
IR_{A+X1}
$IR_{B'/X1}$

(d)

Formability classification

True Positive Rate

0.0
0.2
0.4
0.6
0.8
1.0

False Positive Rate

0.0
0.2
0.4
0.6
0.8
1.0

AUC = 0.95

(e)

Bandgap classification

True Positive Rate

0.0
0.2
0.4
0.6
0.8
1.0

False Positive Rate

0.0
0.2
0.4
0.6
0.8
1.0

AUC = 0.99

(f)

Bandgap regression

Prediction bandgap E_g^{ML} (eV)

0
1
2
3
4
5

Test bandgap E_g^{PBE} (eV)

0
1
2
3
4
5

$R^2 = 0.92$
$MSE = 0.12$ eV